Lecture-15
Conservation Laws for Control Volumes III: Conservation of Linear Momentum

Saikat Mukherjee

Department of Mechanical Engineering, IIT Kharagpur

Learning Objectives

By the end of this lecture, you should be able to

1. Derive the law of **conservation of momentum** from the Reynolds transport equation
2. Identify the **external forces** acting on common systems involving fluids
3. Use the momentum conservation principle for a **bunch of engineering problems**

Conservation of momentum for an inertial CV

- The **rate of change of linear momentum** of a system is equal to the **net force because of surface and body forces** acting on the system

$$\left(\frac{d\vec{P}}{dt}\right)_{\text{system}} = \frac{\partial}{\partial t}\int_{\text{CV}} \vec{V}\,\rho\, d\forall + \int_{\text{CS}} \vec{V}\rho\vec{V}\cdot d\vec{A}$$

$$\vec{F} = \vec{F}_S + \vec{F}_B = \frac{\partial}{\partial t}\int_{\text{CV}} \vec{V}\rho d\forall + \int_{\text{CS}} \vec{V}\rho\vec{V}\cdot d\vec{A}$$

- **The total force due to surface and body forces acting on a control volume leads to a rate of change of momentum within the control volume, and/or a rate at which momentum leaves the control volume through the control surface.**

Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

CV with rectilinear acceleration

- The form of the momentum equation developed in the previous lecture is not valid when the control volume itself has a **rectilinear acceleration**.

$$\vec{F}_S + \vec{F}_B - \int_{CV} \vec{a}_{rf}\, \rho\, d\forall = \frac{\partial}{\partial t} \int_{CV} \vec{V}_{xyz}\, \rho\, d\forall + \int_{CS} \vec{V}_{xyz}\, \rho \vec{V}_{xyz} \cdot d\vec{A}$$

Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Revisiting the learning objectives

1. Identify the **external forces** acting on common systems involving fluids
2. Use the momentum conservation principle for a **bunch of engineering problems**

